CHENHAO ZHANG

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RESEARCH INTERESTS

Vehicle–drone collaborative routing and last-mile logistics; space–time(–state) network modeling; mixed-integer programming and exact decomposition algorithms (Benders decomposition, branch-and-price, branch-and-cut, ADMM); stochastic programming for electrified and crowdsourced delivery systems; urban air mobility logistics.

EDUCATION

Southeast University, Nanjing, China — Mar 2023 – Jun 2027 (expected)
Ph.D. in Transportation Engineering
Advisor: Prof. Lin Cheng

Georgia Institute of Technology, Atlanta, GA, USA — Apr 2026 – Apr 2027
Visiting Ph.D. Scholar (China Scholarship Council award)
Host: Prof. Srinivas Peeta, Editor-in-Chief of *Transportation Research Part B*

Southeast University, Nanjing, China — Sep 2020 – Dec 2022
M.S. in Transportation Engineering (direct admission)
Advisor: Prof. Lin Cheng

Chang'an University, Xi'an, China — Aug 2016 – Jun 2020
B.S. in Traffic Engineering, School of Highway
Advisor: Prof. Hong Chen

PUBLICATIONS

First-authored: **6** SCI journal articles, 3 under review, **7** TRB Annual Meeting papers.
Co-authored: **4** SCI articles (IEEE T-ITS, TRR, etc.), **2** AI conference papers (ICLR, ACL).

Peer-Reviewed Journal Articles (First Author)

[1] **Zhang, C.**, Wu, Y., Feng, T., Zhang, Y., Song, M., Cheng, L. Benders decomposition for traveling salesman problem with drone: A space-time-state network perspective. *Transportation Research Part C: Emerging Technologies*, 2025, 179: 105255

[2] **Zhang, C.**, Xie, L., Wu, Y., Feng, T., Cheng, L. A two-stage stochastic programming approach for the time-dependent vehicle routing problem with stochastic demands: A dynamic space–time network perspective. *Transportation Research Part E: Logistics and Transportation Review*, 2026, 215: 105105

[3] **Zhang, C.**, Feng, T., Wu, Y., Li, Y., Cheng, L. A physical-network-based space-time-state modeling and branch-and-price framework for the time-dependent vehicle routing problem with drones. *Transportation Research Part E: Logistics and Transportation Review*, 2026, accepted for publication

[4] **Zhang, C.**, Du, M., Hu, Z., Jiang, X., Liu, Y., Cheng, L. Benders decomposition for the charging station location-routing problem under stochastic energy consumption: A two-stage model with a space-electricity network perspective. *Applied Energy*, 2026, 416: 127926

[5] **Zhang, C.**, Yong, R., Jiang, X., Zhang, Y., Song, M., Cheng, L. Benders decomposition for charging station location-routing problem with time-dependent travel times: A space-time-electricity network perspective. *Energy*, 2025, 333: 137310

[6] **Zhang, C.**, Cheng, L., Zhang, Y., Du, M., Zheng, W. Route optimization of vacant taxicab considering sequential dependence in abstract grid network based on quadtree. *Transportation Letters*, 2024, 16: 338–353

Manuscripts Under Review (First Author)

[7] **Zhang, C.**, Jiang, Z., Song, M., Du, M., Cheng, L. Branch-and-cut algorithm for the time-dependent flying sidekick traveling salesman problem with mid-route rendezvous on a space-time network, under review

[8] **Zhang, C.**, Kibria, M. G., Liu, Y., Yang, R., Lu, S., Peeta, S., Cheng, L. Stochastic programming approach for joint transfer-station location and dedicated-crowdsourced delivery planning, under review

[9] **Zhang, C.**, Li, Y., Liu, H., Zhang, Q., Cheng, L. Vehicle routing problem with multiple trucks and drones considering time-dependent no-fly zones and payload-dependent energy consumption, under review

Conference Presentations

[10] Seven papers (6 first-authored, 1 corresponding-authored) presented at the 104th–106th *TRB Annual Meetings*, Washington, D.C., 2025–2027.

RESEARCH GRANT EXPERIENCE

- **National Natural Science Foundation of China (NSFC) General Program (52172318)** 2022–2025
  — Urban network design via individual space–time accessibility.
  Named participant; developed space–time network models and decomposition algorithms.
- **Natural Science Foundation of Jiangsu Province (BK20221469)** 2022–2025
  — Urban network equilibrium with turning flows.
  Named participant; network data processing, parameter calibration, numerical validation.
- **NSFC Key Program (52131203)** 2022–2026
  — Big-data-driven traffic flow modeling and control.
  Processed multi-source data; built flow-based network models.
- **Industry projects** 2020–2022
  — Traffic impact analyses and county-level transportation data resource development.
  Led 5 projects; completed two end-to-end (surveys, TransCAD modeling, reports).

TEACHING EXPERIENCE

- **Guest Lecturer (English-medium)** — Delivered a 5-session (10 hours) linear programming module for international Ph.D. students, Southeast University.
- **Teaching Assistant** — *Traffic Engineering* (undergraduate) and *Urban Transportation Network Analysis* (graduate, 2 years); led problem sessions, graded assignments and exams.
- **Open Online Courses** — Public video series on LP/MILP modeling and decomposition algorithms with logistics case studies; 800,000+ views, 11,000+ subscribers.
- **Learning Community** — Founded the MIP Operations Research community: ~700 members, 100+ universities, 20+ countries; regular research seminars.

PROFESSIONAL SERVICE & INTERNATIONAL COLLABORATION

- **Reviewer** — *Transportation Research Part E*; *Automation in Construction*; Sustainable Energy, Grids and Networks; *Journal of Heuristics*; *Transportation Letters*; *Multimodal Transportation*; *TRB Annual Meeting*.
- **International Collaborators** — Prof. Srinivas Peeta (Georgia Tech; Editor-in-Chief, *TR Part B*); Prof. Lin Xie (BTU Cottbus-Senftenberg, Germany; Associate Editor, *BISE*); Asst. Prof. Yaoxin Wu (Eindhoven University of Technology, Netherlands).
- **Open Source** — Public implementations of MILP decomposition algorithms (Benders, Lagrangian relaxation, branch-and-bound/cut); 140+ GitHub stars.
- **Science Communication** — Section editor (transportation & logistics), leading Chinese OR outreach platform; 30,000+ article reads.

PATENTS & SOFTWARE

Nine granted Chinese invention patents and four registered software copyrights (first holder), covering vehicle routing, taxi dispatch, and metro passenger flow management.

SELECTED HONORS & AWARDS

China Scholarship Council (CSC) Joint Ph.D. Scholarship (Georgia Tech)	2025
National Second Prize, China Post-Graduate Mathematical Contest in Modeling (team leader)	2023–2025
Graduate Student of the Year, Southeast University (10 students university-wide)	2025
First-Class Academic Scholarship, Southeast University (top 10%)	2024
Growth Fund, School of Transportation (3 students school-wide)	2024

TECHNICAL SKILLS

Optimization: Gurobi/CPLEX; Benders, branch-and-price/cut, Lagrangian relaxation, ADMM. **Programming:** Python. **Tools:** TransCAD, SUMO, ArcGIS. **Languages:** Mandarin (native), English (fluent).